

MODULE 5.1 · Concept 8 of 8 · CODING LAB

Building a Simple Agent

Hands-on: your first LangChain ReAct agent with tool calling — 20-minute lab

20 min

Duration

B3 – Apply

Bloom's

L2

Level

TH-000006

Prerequisite

AI-AG-FN-CD-000001

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SECTION 1 — WHY & WHAT

From Theory to Code

Every concept you learned becomes 20 lines of Python

Step 1: Define Tools and Create the Agent

Three components: tools (actions), LLM (brain), agent (loop)

```
Python — agent_setup.py
from langchain.agents import create_react_agent
from langchain_openai import ChatOpenAI
from langchain.tools import Tool

# Define tools the agent can use
def web_search(query: str) -> str:
    """Search the web for info"""
    return search_api(query) # your impl

def calculator(expr: str) -> str:
    """Evaluate a math expression"""
    return str(eval(expr)) # safe eval

tools = [
    Tool(name="search",
        func=web_search,
        description="Search the web"),
    Tool(name="calc",
        func=calculator,
        description="Do math"),
]

# Create agent: LLM brain + tools
llm = ChatOpenAI(model="gpt-4o-mini")
agent = create_react_agent(llm, tools)
```

Theory → Code Mapping

Action	→ Tool(func=...)
Perception	→ Tool result → context
Reasoning	→ ChatOpenAI (LLM)
Memory	→ Context window (auto)
Goal	→ agent.invoke("...")
Loop	→ create_react_agent()

Step 2: Run the Agent and Read the Trace

The agent loop in action — Thought → Action → Observation → repeat

```
# Run the agent with a multi-step query
Python — run_agent.py

result = agent.invoke(

    "What is the population of Tokyo

    divided by 3?"
```

Agent Trace Output

- 1 **Thought:** I need Tokyo's population. Let me search.
- 2 **Action:** search("Tokyo population 2024")
- 3 **Observation:** 13.96 million (2024 census)
- 4 **Thought:** Now I need to divide by 3.
- 5 **Action:** calc("13960000 / 3")
- 6 **Observation:** 4653333.33
- 7 **Answer:** ≈ 4.65 million

Token Count: 7 steps × ~80 tokens avg = 560 tokens. At GPT-4o-mini (\$0.15/1M in + \$0.60/1M out): cost ≈ \$0.0004 per query. A human researcher would take 2 minutes for the same task.

Lab Exercises — Try These Yourself

Extend the basic agent with new tools and harder queries

Exercise 1

Easy

Add a `weather_lookup` tool and ask: 'Should I carry an umbrella in Bangalore today?'

Exercise 2

Medium

Add a `code_executor` tool and ask: 'Generate and run a Python script to find the first 20 primes'

Exercise 3

Medium

Add error handling: what happens if the search tool returns 'No results found'? Implement a retry with reformulated query.

Exercise 4

Advanced

Add trace logging: print each Thought/Action/Observation to a file. Count tokens per step. Plot token usage.

Key Takeaways

Three components to code: tools (Tool class), brain (ChatOpenAI), loop (create_react_agent).

Theory → code mapping is direct: Perception = tool results, Reasoning = LLM, Action = tool calls.

The agent trace is your debugging tool — every Thought/Action/Observation is logged and inspectable.

Cost per query: ~\$0.0004 for a multi-step reasoning task with GPT-4o-mini.

This coding pattern is the foundation for everything in Modules 5.2–5.7 and Module 9.

AI-AG-FN-CD-000001 Complete

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Concept 8 of 8 — Coding Lab

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